

VALDEZ WSO, AK, USA

WMO: 702750

Lat: 61.130N

Lon: 146.352W

Elev: 29

StdP: 100.98

Time Zone: -9.00 (NAK)

Period: 94-05

WBAN: 26442

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
1	-15.3	-13.6	-25.6	0.4	-12.9	-23.7	0.4	-12.6	14.6	-7.8	13.1	-7.1	6.1	60	0.690

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	7.1	21.5	13.6	19.9	13.3	18.3	12.8	14.9	19.9	14.1	18.6	13.4	17.1	4.2	240

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
12.7	9.2	15.6	12.1	8.8	14.7	11.5	8.5	14.1	41.5	20.2	39.4	18.8	37.5	17.2	17.3

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max		
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(4) 11.6	8.8	7.4	DB	-17.2	26.2	2.3	2.0	-18.9	27.6	-20.2	28.8	-21.5	29.9	-23.1	31.4	(4)
(5)			WB	-18.3	16.0	2.1	0.8	-19.8	16.6	-21.0	17.1	-22.2	17.6	-23.7	18.2	(5)

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec		
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(6) Temperatures, Degree-Days and Degree-Hours	DBAvg	4.2	-4.6	-2.7	-0.8	3.8	8.5	12.4	13.2	12.5	8.8	3.8	-1.3	-3.4	(6)	
	DBStd	7.19	4.83	4.19	3.33	2.80	2.56	2.26	1.85	2.11	2.27	2.94	3.52	4.14	(7)	
	HDD10.0	2391	453	357	336	186	60	3	0	2	47	193	338	416	(8)	
	HDD18.3	5152	712	590	594	434	306	178	159	180	285	451	588	674	(9)	
	CDD10.0	282	0	0	0	1	13	76	99	81	12	0	0	0	(10)	
	CDD18.3	1	0	0	0	0	0	1	1	0	0	0	0	0	(11)	
	CDH23.3	22	0	0	0	0	0	7	9	6	0	0	0	0	(12)	
	CDH26.7	1	0	0	0	0	0	1	1	0	0	0	0	0	(13)	
(14) Wind	WSAvg	2.7	3.5	2.9	3.3	2.3	2.6	2.7	2.1	1.9	1.9	2.6	3.2	3.3	(14)	
(15) Precipitation	PrecAvg	1673	150	142	112	88	75	70	93	163	221	211	144	182	(15)	
	PrecMax	2370	386	361	254	206	169	154	228	463	424	392	523	440	(16)	
	PrecMin	884	0	14	21	14	18	3	5	9	18	8	11	34	(17)	
	PrecStd	325	96	87	67	51	35	35	48	98	99	101	109	109	(18)	
(19) Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	6.0	4.1	7.9	16.7	20.5	24.2	24.5	24.1	20.9	11.9	6.7	4.8	(19)	
		MCWB	1.9	1.2	2.5	9.0	10.0	14.2	14.6	14.1	11.9	6.0	4.4	1.5	(20)	
	2%	DB	2.8	3.1	5.8	12.7	17.7	21.4	20.7	20.7	16.7	10.1	5.3	3.4	(21)	
		MCWB	0.9	0.8	1.7	6.6	9.8	13.6	14.4	13.9	10.2	5.4	4.0	1.7	(22)	
	5%	DB	1.7	2.3	4.4	10.1	15.3	19.5	19.0	18.9	14.4	8.9	4.1	2.2	(23)	
		MCWB	0.4	0.6	1.2	5.2	8.9	13.1	13.9	13.5	9.4	5.3	1.9	0.8	(24)	
	10%	DB	1.0	1.7	3.3	8.4	13.6	17.5	17.4	17.2	12.4	7.8	2.9	1.3	(25)	
		MCWB	0.0	0.4	0.7	4.4	8.3	12.2	13.3	12.8	9.0	5.0	1.3	0.4	(26)	
(27) Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	2.9	2.4	3.9	9.2	11.6	15.7	15.7	15.6	12.3	8.9	5.7	3.5	(27)	
		MCDB	4.8	3.0	5.7	16.0	17.6	22.1	21.1	20.4	17.2	9.2	6.0	3.9	(28)	
	2%	WB	1.5	1.8	2.8	7.1	10.4	14.5	14.9	14.7	11.3	7.6	4.7	2.2	(29)	
		MCDB	2.1	2.3	4.5	12.0	15.7	19.9	19.8	19.3	14.7	8.3	5.1	2.7	(30)	
	5%	WB	0.8	1.3	2.0	5.7	9.8	13.5	14.1	13.9	10.6	6.7	2.8	1.3	(31)	
		MCDB	1.3	1.8	3.5	9.1	14.3	18.3	18.5	18.0	12.7	7.6	3.4	1.7	(32)	
	10%	WB	0.3	0.7	1.3	4.9	9.0	12.6	13.4	13.0	10.0	5.9	1.7	0.7	(33)	
		MCDB	0.8	1.2	2.6	7.6	12.8	16.8	16.8	16.3	11.7	6.9	2.4	1.1	(34)	
(35) Mean Daily Temperature Range	5% DB	MDBR	4.1	4.6	5.5	6.8	7.8	8.2	7.1	7.7	6.5	4.9	4.0	4.0	(35)	
		MCDBR	4.6	4.8	6.8	10.1	12.0	12.7	11.1	11.9	10.7	6.4	4.7	4.3	(36)	
	5% WB	MCWBR	3.4	3.5	4.6	5.7	6.2	6.1	5.7	6.1	5.7	3.9	3.1	3.0	(37)	
		MCWBR	3.8	3.6	5.4	8.7	9.9	11.4	9.8	10.5	7.6	4.7	3.3	3.4	(38)	
(40) Clear-Sky Solar Irradiance	taub	taub	0.259	0.318	0.325	0.386	0.402	0.404	0.404	0.401	0.343	0.319	0.273	0.251	(40)	
		taud	2.006	2.026	2.065	1.940	2.039	2.165	2.230	2.235	2.362	2.188	1.984	1.895	(41)	
	Edn at Noon	Edn at Noon	433	617	773	786	811	820	810	776	766	623	390	248	(42)	
		Edh at Noon	43	81	109	151	148	133	122	112	82	68	43	29	(43)	
(44) All-Sky Solar Radiation	RadAvg	RadAvg	0.28	0.99	2.57	4.38	5.46	5.66	5.07	3.85	2.48	1.14	0.41	0.15	(44)	
	RadStd	RadStd	0.03	0.13	0.24	0.34	0.38	0.32	0.31	0.36	0.32	0.13	0.06	0.01	(45)	

Historical Trends

		DBAvg	Heating		Cooling			Degree-Days			
			99% DB	99% DP	1% DB	1% WB	1% DP	HDD10.0	HDD18.3	CDD10.0	CDD18.3
(46)	Station Only	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)
(47)	Regional (21 neighbors)	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A

Nomenclature: See separate page